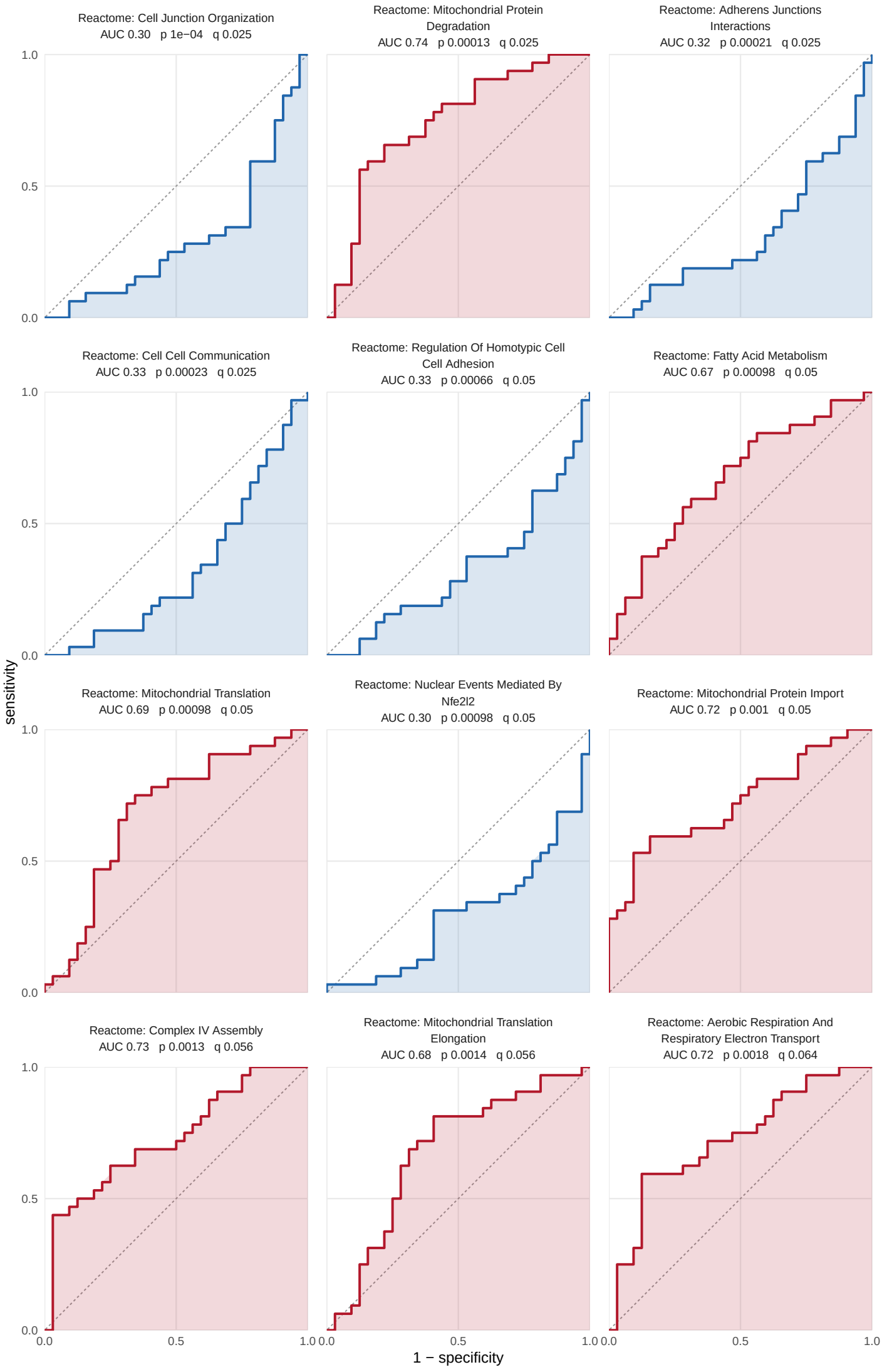
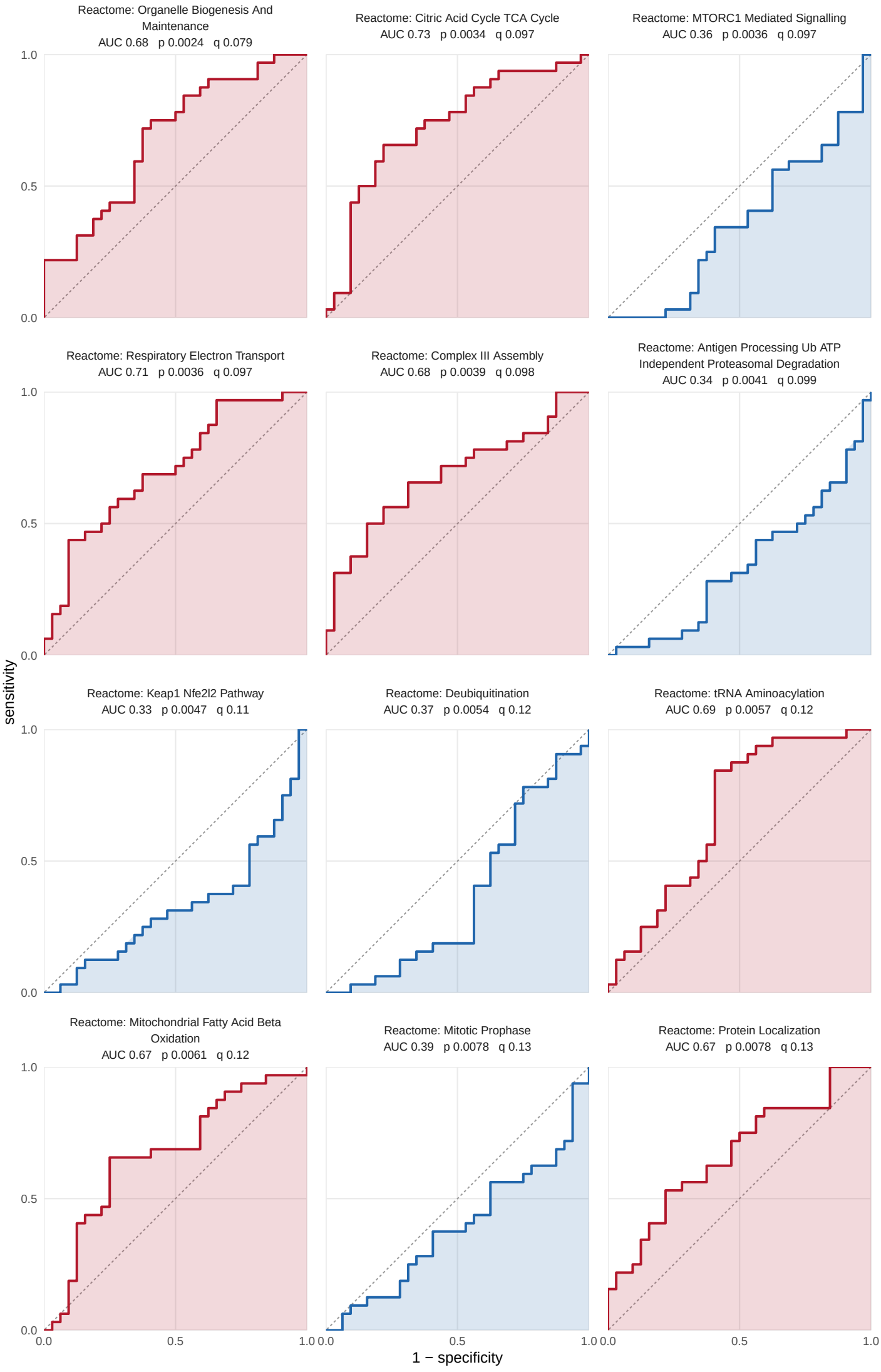
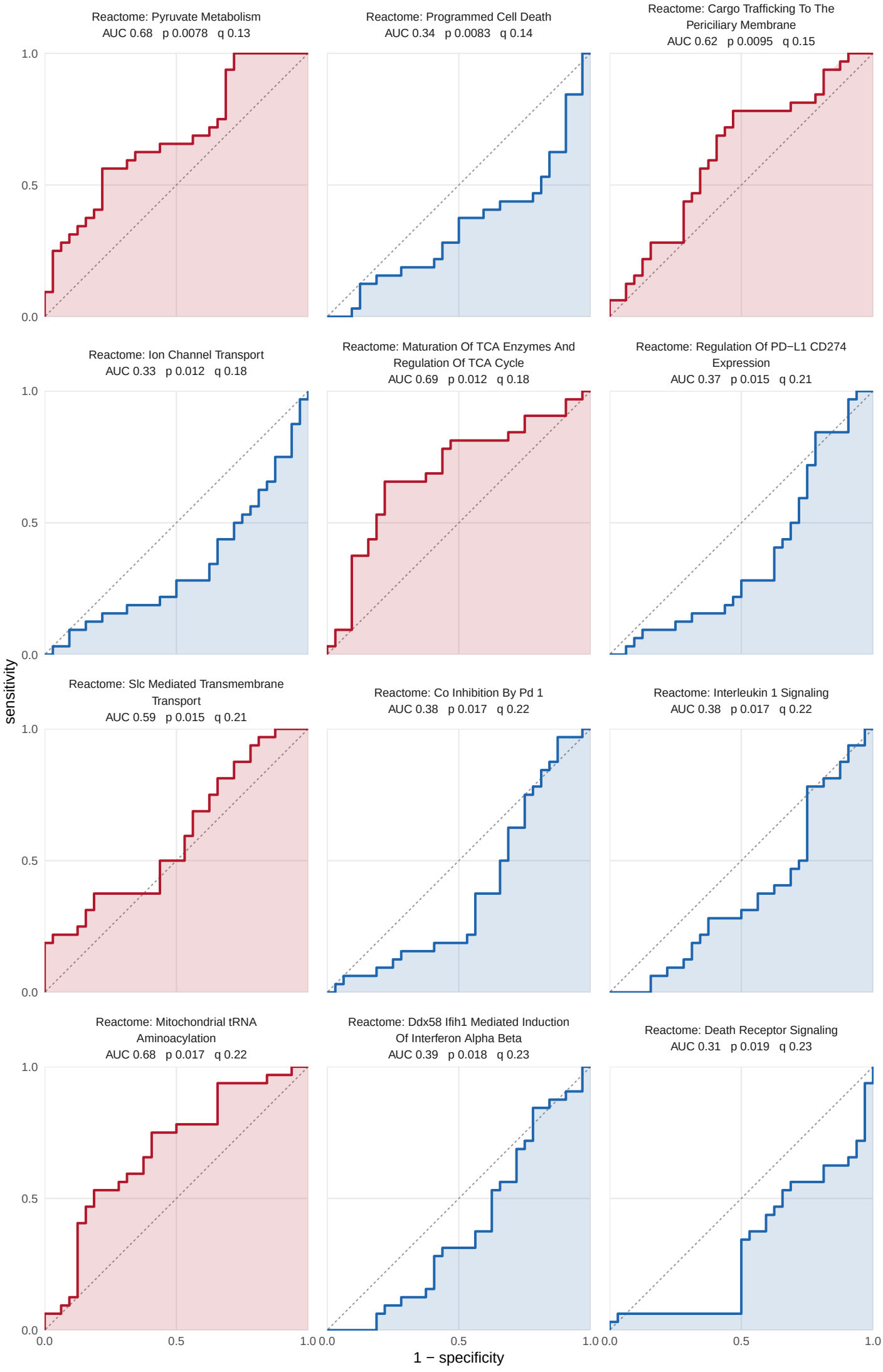
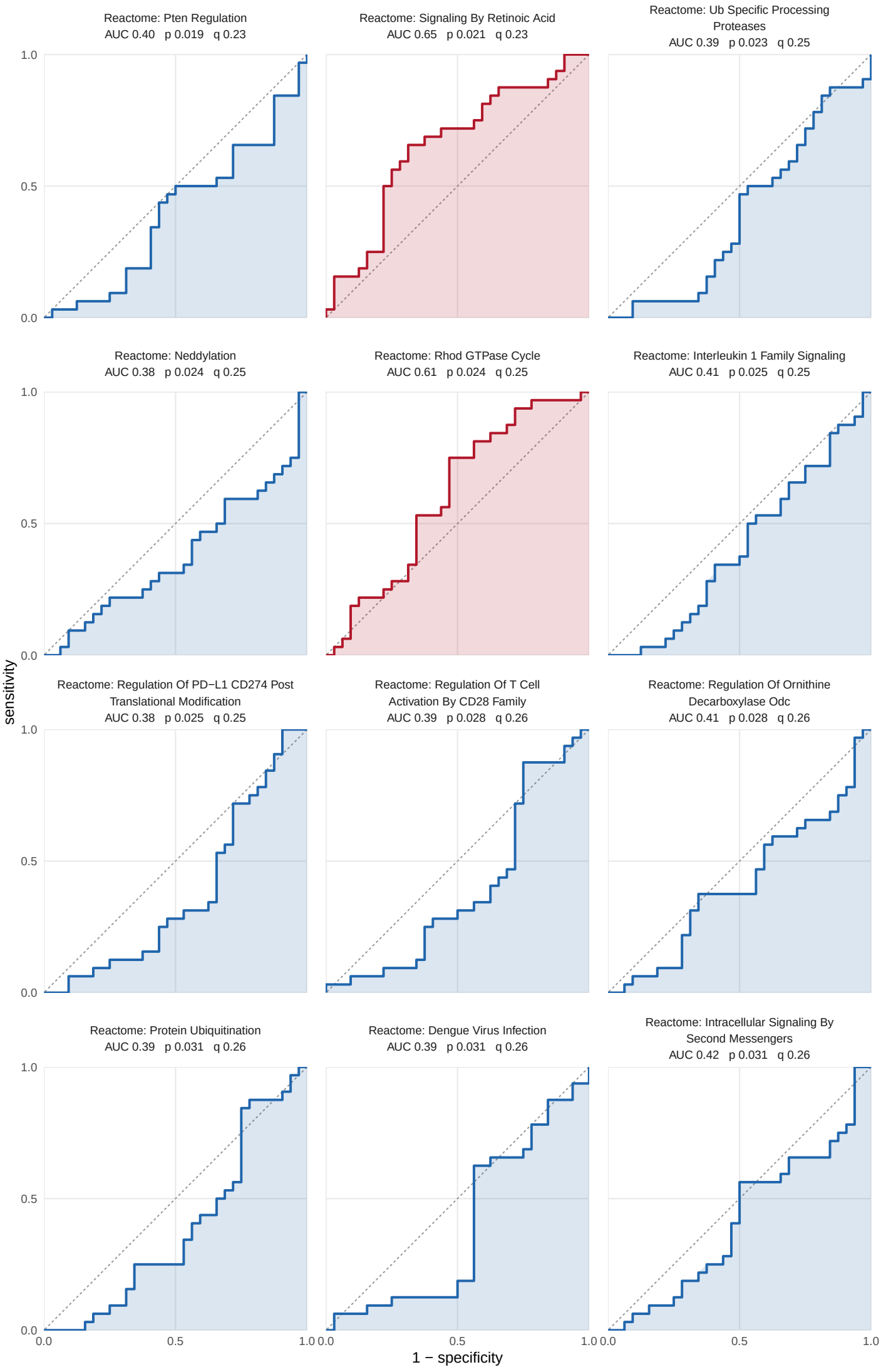
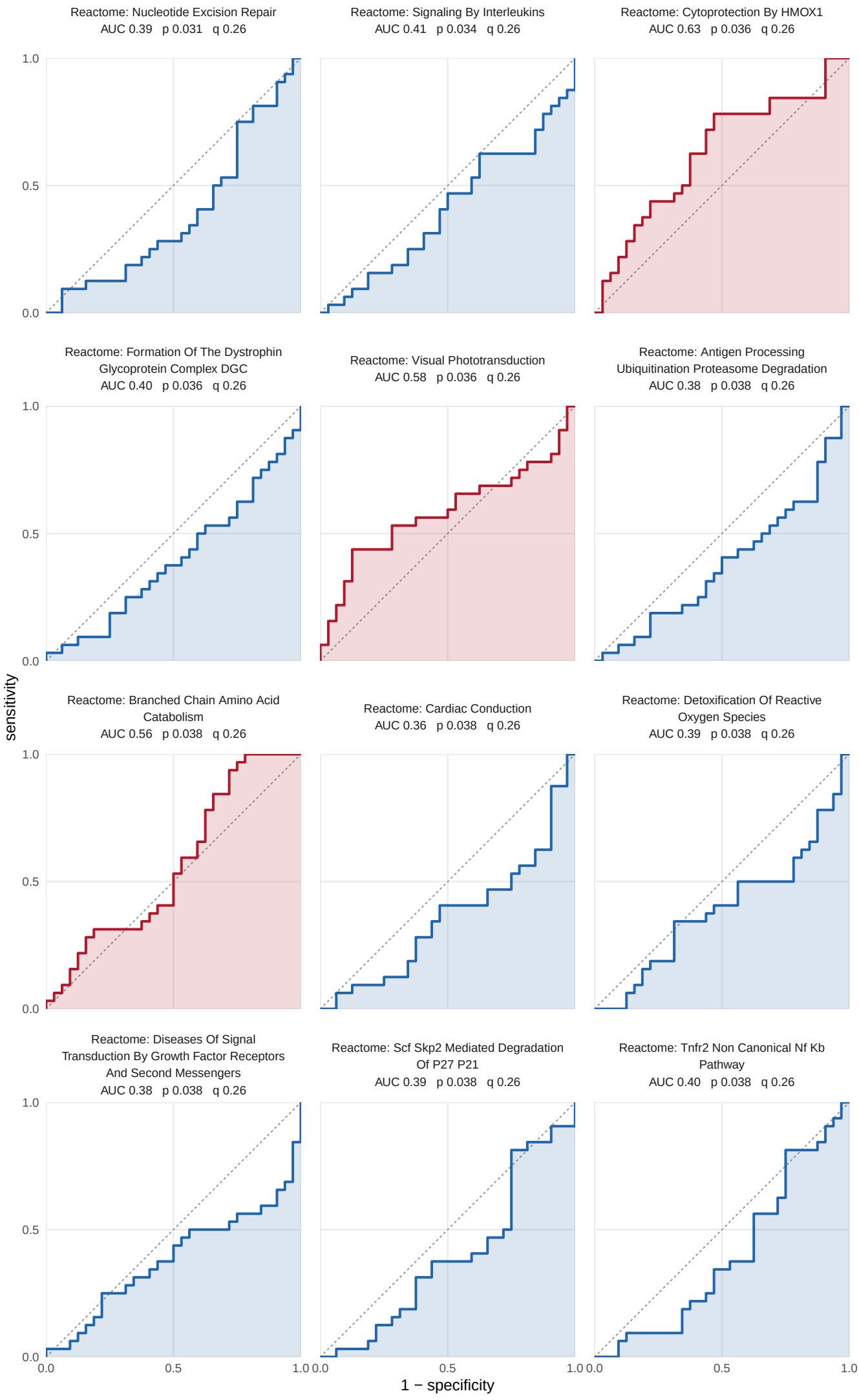
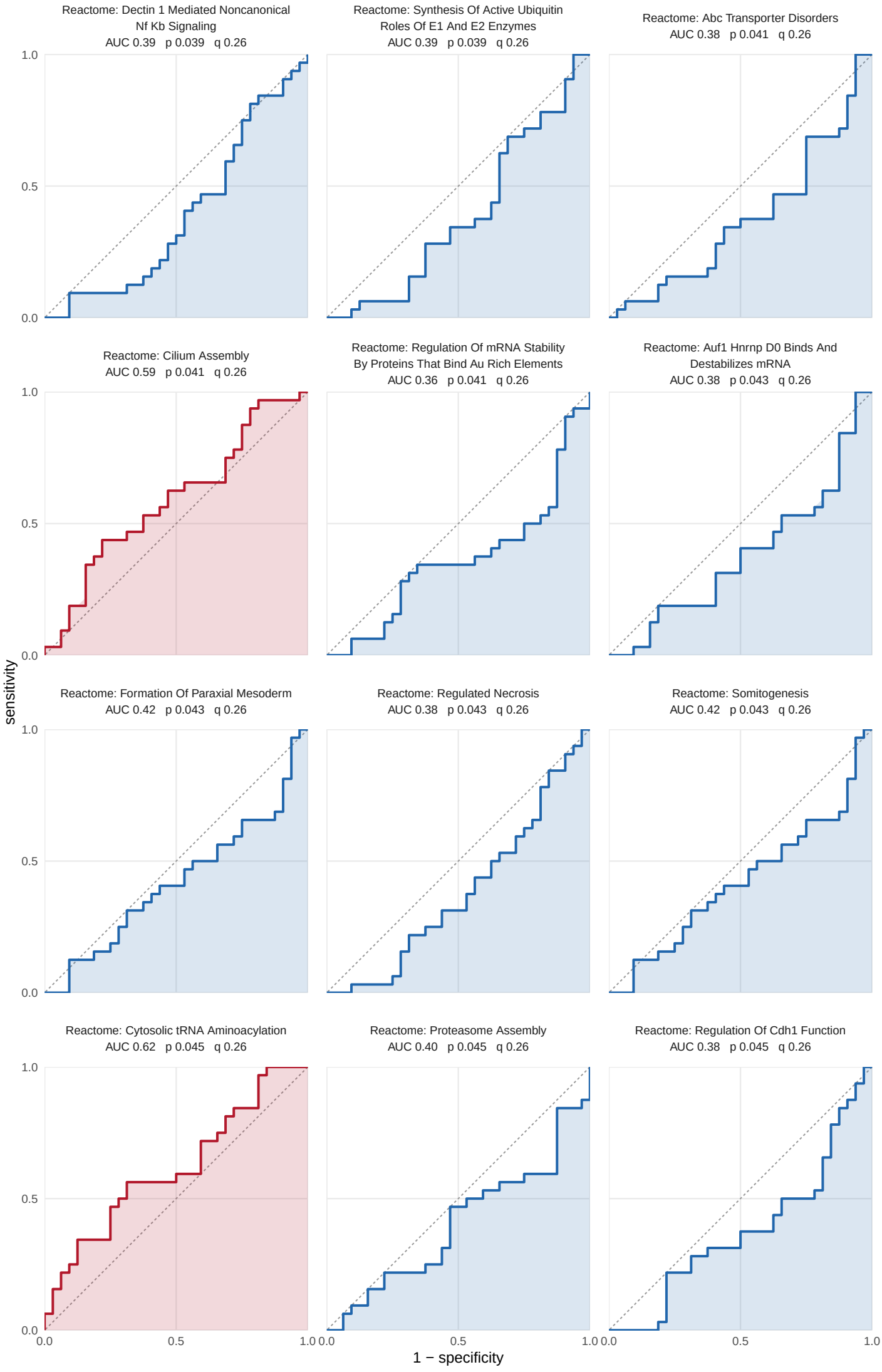

Reactome: Cell Junction Organization
AUC 0.30 p 1e−04 q 0.025
Reactome: Mitochondrial Protein Degradation
AUC 0.74 p 0.00013 q 0.025
Reactome: Adherens Junctions Interactions
AUC 0.32 p 0.00021 q 0.025
Reactome: Cell Cell Communication
AUC 0.33 p 0.00023 q 0.025
Reactome: Regulation Of Homotypic Cell Cell Adhesion
AUC 0.33 p 0.00066 q 0.05
Reactome: Fatty Acid Metabolism
AUC 0.67 p 0.00098 q 0.05
Reactome: Mitochondrial Translation
AUC 0.69 p 0.00098 q 0.05
Reactome: Nuclear Events Mediated By Nfe2l2
AUC 0.30 p 0.00098 q 0.05
Reactome: Mitochondrial Protein Import
AUC 0.72 p 0.001 q 0.05
Reactome: Complex IV Assembly
AUC 0.73 p 0.0013 q 0.056
Reactome: Mitochondrial Translation Elongation
AUC 0.68 p 0.0014 q 0.056
Reactome: Aerobic Respiration And Respiratory Electron Transport
AUC 0.72 p 0.0018 q 0.064
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Organelle Biogenesis And Maintenance
AUC 0.68 p 0.0024 q 0.079
Reactome: Citric Acid Cycle TCA Cycle
AUC 0.73 p 0.0034 q 0.097
Reactome: MTORC1 Mediated Signalling
AUC 0.36 p 0.0036 q 0.097
Reactome: Respiratory Electron Transport
AUC 0.71 p 0.0036 q 0.097
Reactome: Complex III Assembly
AUC 0.68 p 0.0039 q 0.098
Reactome: Antigen Processing Ub ATP Independent Proteasomal Degradation
AUC 0.34 p 0.0041 q 0.099
Reactome: Keap1 Nfe2l2 Pathway
AUC 0.33 p 0.0047 q 0.11
Reactome: Deubiquitination
AUC 0.37 p 0.0054 q 0.12
Reactome: tRNA Aminoacylation
AUC 0.69 p 0.0057 q 0.12
Reactome: Mitochondrial Fatty Acid Beta Oxidation
AUC 0.67 p 0.0061 q 0.12
Reactome: Mitotic Prophase
AUC 0.39 p 0.0078 q 0.13
Reactome: Protein Localization
AUC 0.67 p 0.0078 q 0.13
sensitivity
1 − specificity
0.0
0.5
1.0

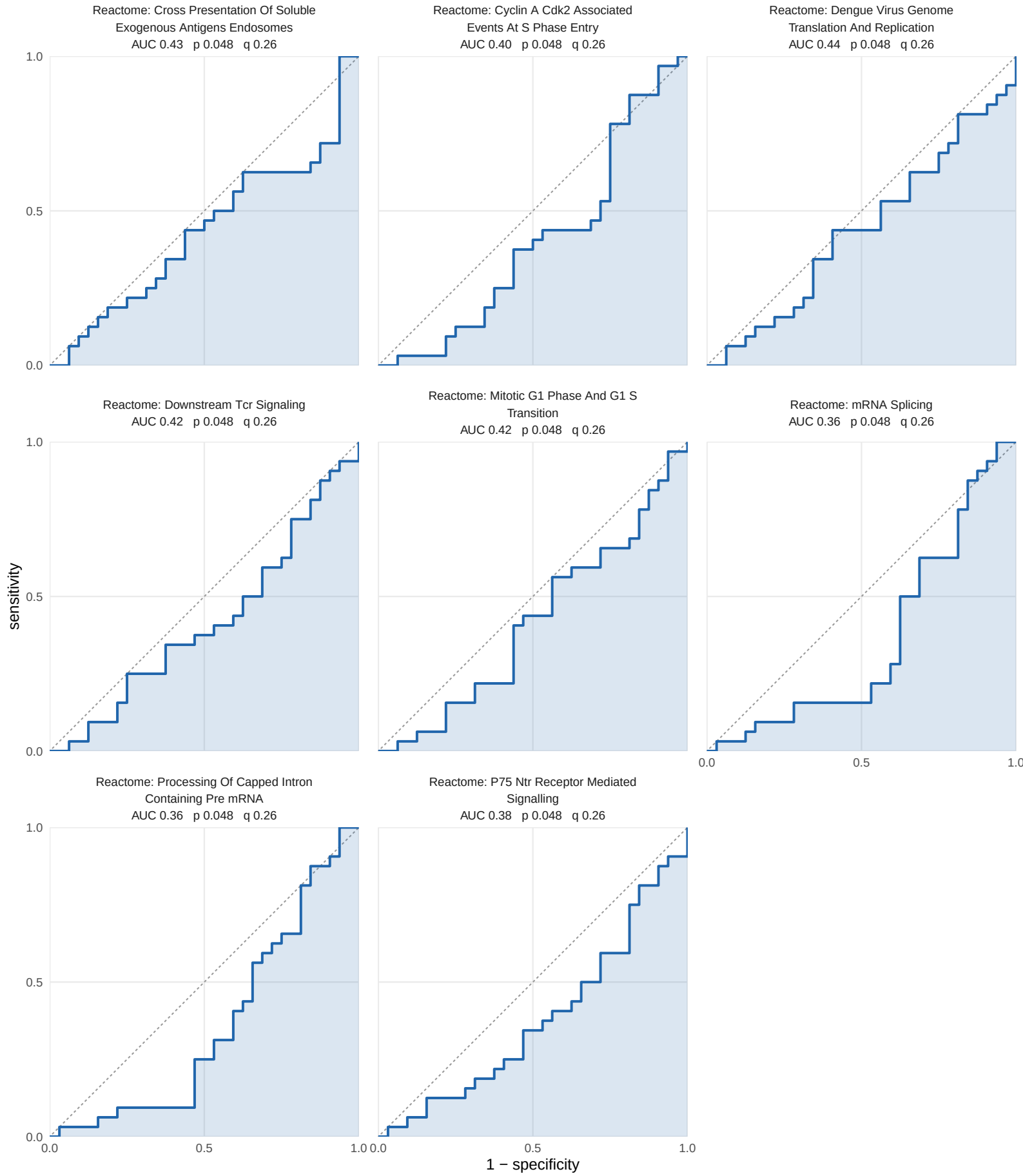

Reactome: Pyruvate Metabolism
AUC 0.68 p 0.0078 q 0.13
Reactome: Programmed Cell Death
AUC 0.34 p 0.0083 q 0.14
Reactome: Cargo Trafficking To The Periciliary Membrane
AUC 0.62 p 0.0095 q 0.15
Reactome: Ion Channel Transport
AUC 0.33 p 0.012 q 0.18
Reactome: Maturation Of TCA Enzymes And Regulation Of TCA Cycle
AUC 0.69 p 0.012 q 0.18
Reactome: Regulation Of PD−L1 CD274 Expression
AUC 0.37 p 0.015 q 0.21
Reactome: Slc Mediated Transmembrane Transport
AUC 0.59 p 0.015 q 0.21
Reactome: Co Inhibition By Pd 1
AUC 0.38 p 0.017 q 0.22
Reactome: Interleukin 1 Signaling
AUC 0.38 p 0.017 q 0.22
Reactome: Mitochondrial tRNA Aminoacylation
AUC 0.68 p 0.017 q 0.22
Reactome: Ddx58 Ifih1 Mediated Induction Of Interferon Alpha Beta
AUC 0.39 p 0.018 q 0.23
Reactome: Death Receptor Signaling
AUC 0.31 p 0.019 q 0.23
sensitivity
1 − specificity
0.0
0.5
1.0

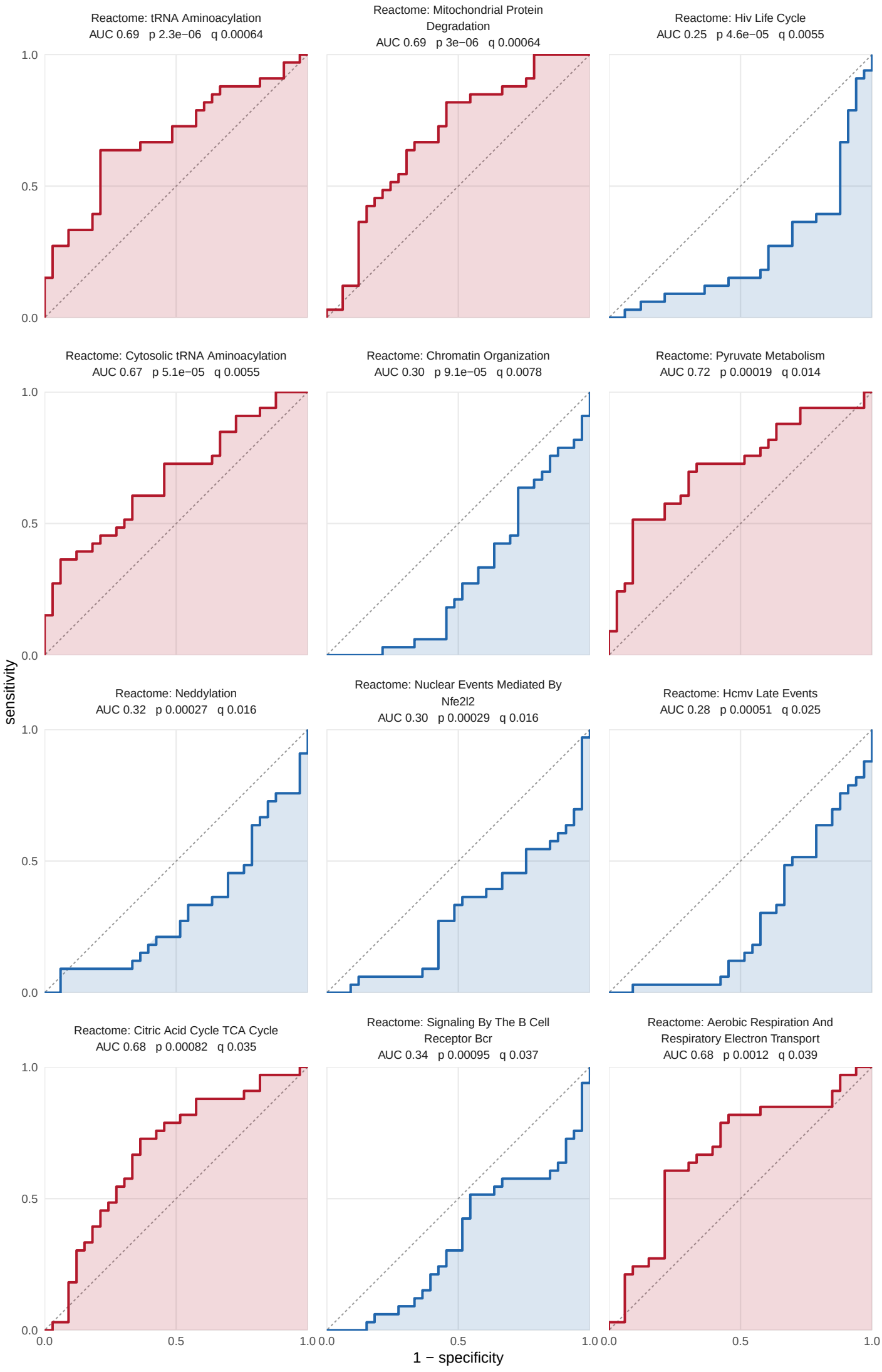

Reactome: Pten Regulation
AUC 0.40 p 0.019 q 0.23
Reactome: Signaling By Retinoic Acid
AUC 0.65 p 0.021 q 0.23
Reactome: Ub Specific Processing Proteases
AUC 0.39 p 0.023 q 0.25
Reactome: Neddylation
AUC 0.38 p 0.024 q 0.25
Reactome: Rhod GTPase Cycle
AUC 0.61 p 0.024 q 0.25
Reactome: Interleukin 1 Family Signaling
AUC 0.41 p 0.025 q 0.25
Reactome: Regulation Of PD−L1 CD274 Post Translational Modification
AUC 0.38 p 0.025 q 0.25
Reactome: Regulation Of T Cell Activation By CD28 Family
AUC 0.39 p 0.028 q 0.26
Reactome: Regulation Of Ornithine Decarboxylase Odc
AUC 0.41 p 0.028 q 0.26
Reactome: Protein Ubiquitination
AUC 0.39 p 0.031 q 0.26
Reactome: Dengue Virus Infection
AUC 0.39 p 0.031 q 0.26
Reactome: Intracellular Signaling By Second Messengers
AUC 0.42 p 0.031 q 0.26
sensitivity
1 − specificity
0.0
0.5
1.0

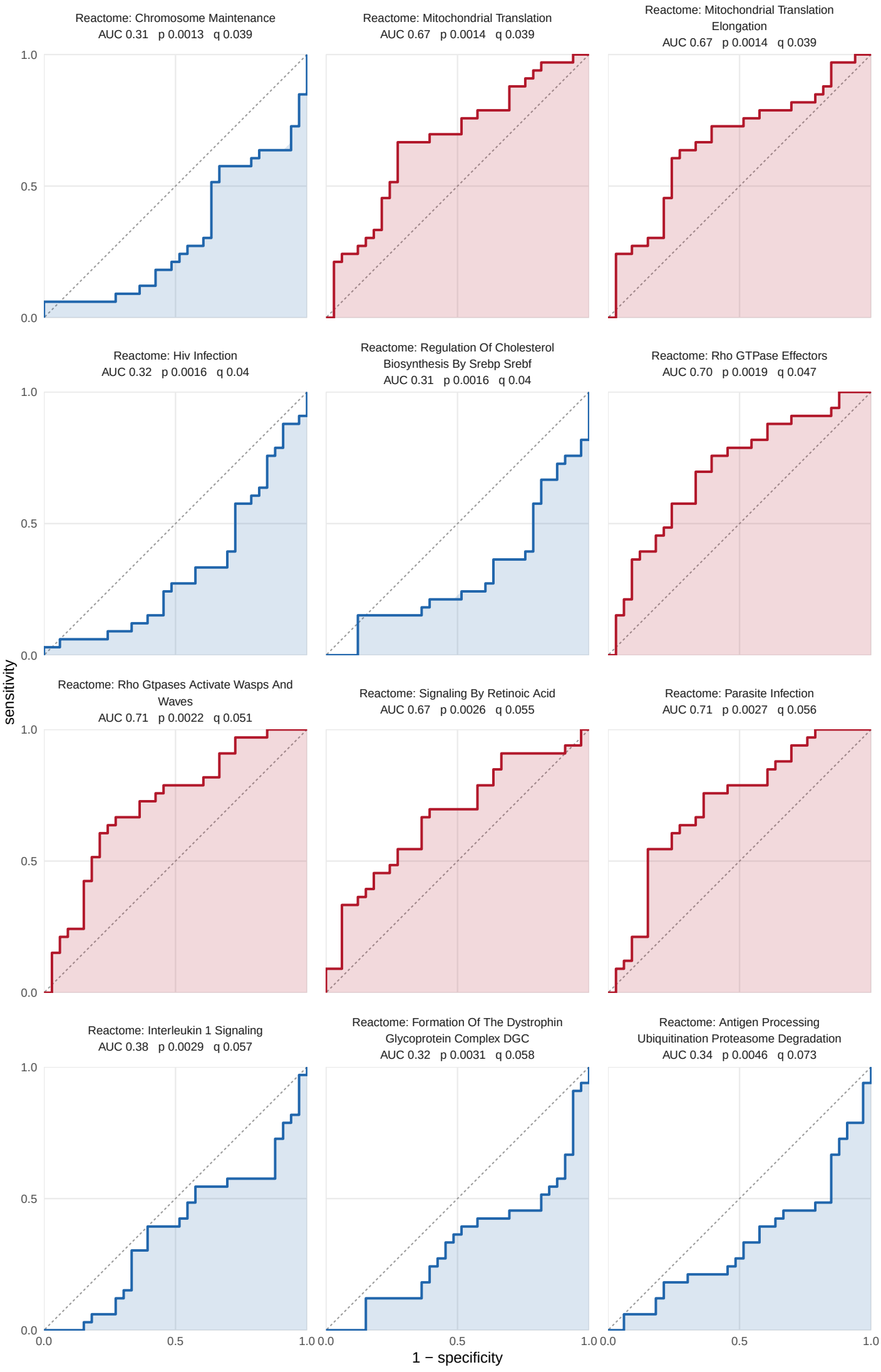

Reactome: Nucleotide Excision Repair
AUC 0.39 p 0.031 q 0.26
Reactome: Signaling By Interleukins
AUC 0.41 p 0.034 q 0.26
Reactome: Cytoprotection By HMOX1
AUC 0.63 p 0.036 q 0.26
Reactome: Formation Of The Dystrophin Glycoprotein Complex DGC
AUC 0.40 p 0.036 q 0.26
Reactome: Visual Phototransduction
AUC 0.58 p 0.036 q 0.26
Reactome: Antigen Processing Ubiquitination Proteasome Degradation
AUC 0.38 p 0.038 q 0.26
Reactome: Branched Chain Amino Acid Catabolism
AUC 0.56 p 0.038 q 0.26
Reactome: Cardiac Conduction
AUC 0.36 p 0.038 q 0.26
Reactome: Detoxification Of Reactive Oxygen Species
AUC 0.39 p 0.038 q 0.26
Reactome: Diseases Of Signal Transduction By Growth Factor Receptors And Second Messengers
AUC 0.38 p 0.038 q 0.26
Reactome: Scf Skp2 Mediated Degradation Of P27 P21
AUC 0.39 p 0.038 q 0.26
Reactome: Tnfr2 Non Canonical Nf Kb Pathway
AUC 0.40 p 0.038 q 0.26
sensitivity
1 − specificity
0.0
0.5
1.0

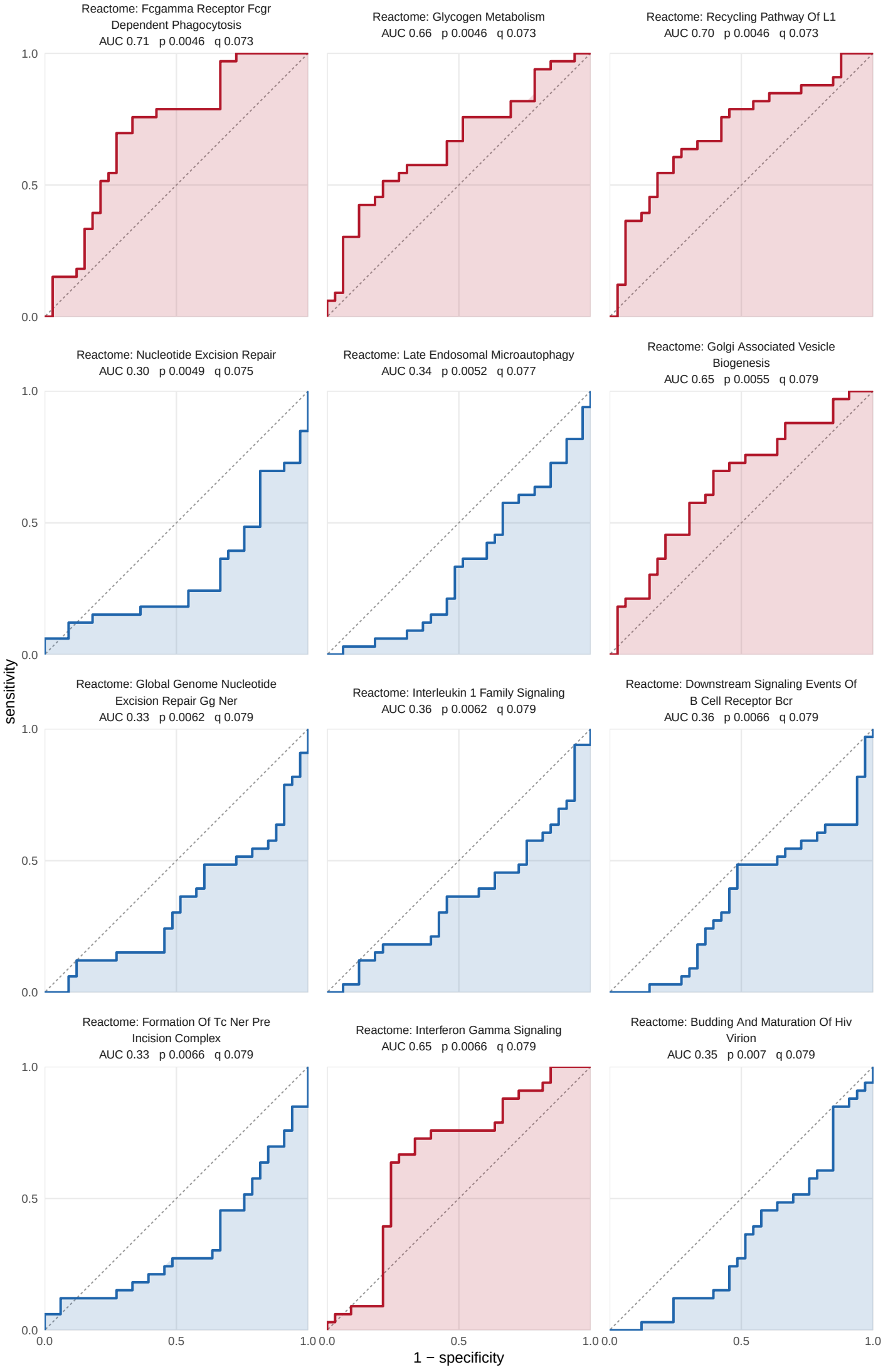

Reactome: Dectin 1 Mediated Noncanonical Nf Kb Signaling
AUC 0.39 p 0.039 q 0.26
Reactome: Synthesis Of Active Ubiquitin Roles Of E1 And E2 Enzymes
AUC 0.39 p 0.039 q 0.26
Reactome: Abc Transporter Disorders
AUC 0.38 p 0.041 q 0.26
Reactome: Cilium Assembly
AUC 0.59 p 0.041 q 0.26
Reactome: Regulation Of mRNA Stability By Proteins That Bind Au Rich Elements
AUC 0.36 p 0.041 q 0.26
Reactome: Auf1 Hnrnp D0 Binds And Destabilizes mRNA
AUC 0.38 p 0.043 q 0.26
Reactome: Formation Of Paraxial Mesoderm
AUC 0.42 p 0.043 q 0.26
Reactome: Regulated Necrosis
AUC 0.38 p 0.043 q 0.26
Reactome: Somitogenesis
AUC 0.42 p 0.043 q 0.26
Reactome: Cytosolic tRNA Aminoacylation
AUC 0.62 p 0.045 q 0.26
Reactome: Proteasome Assembly
AUC 0.40 p 0.045 q 0.26
Reactome: Regulation Of Cdh1 Function
AUC 0.38 p 0.045 q 0.26
sensitivity
1 − specificity
0.0
0.5
1.0

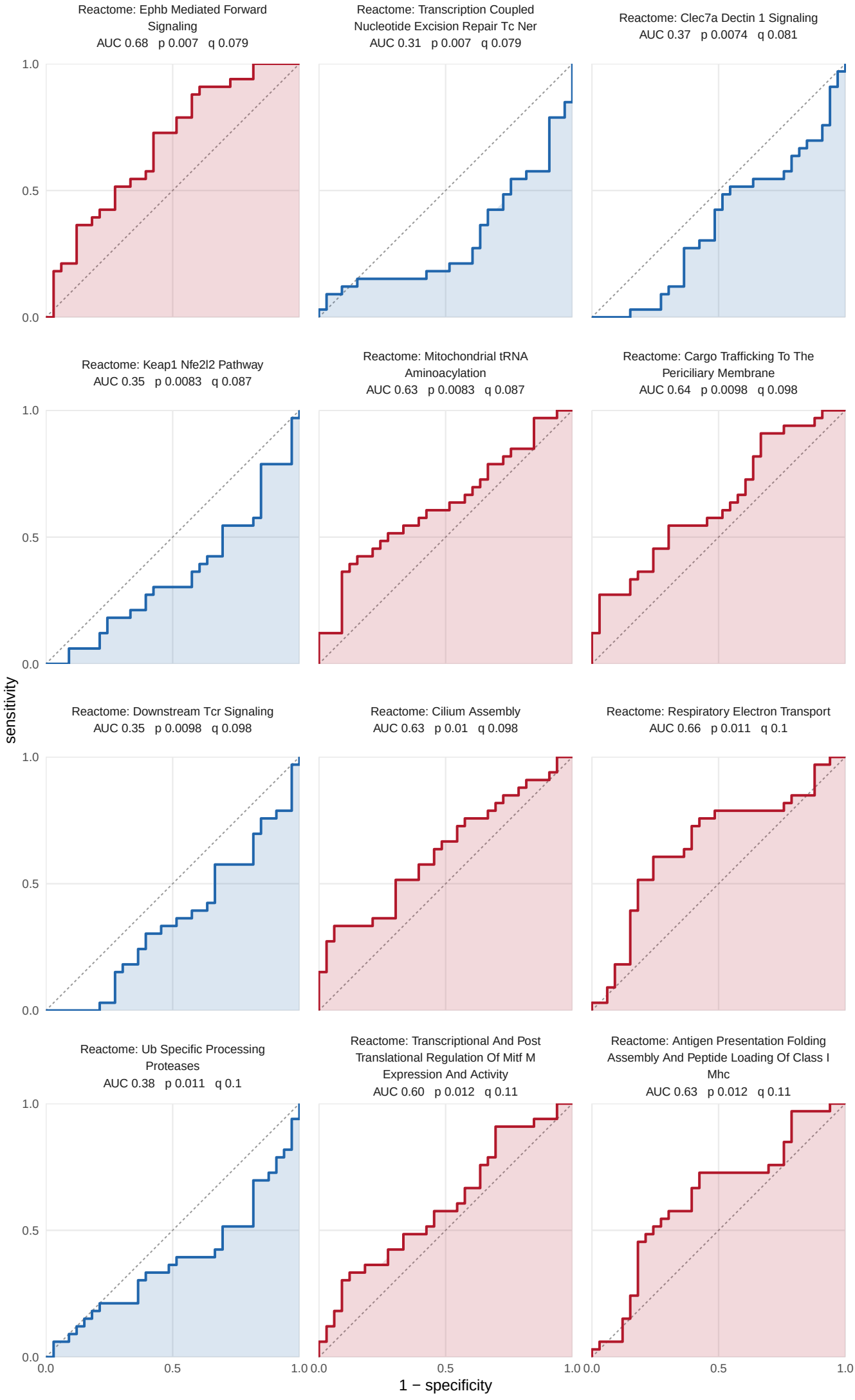
S6. ROC curves, Pre to post, BFR, Reactome (page 7 of 7)

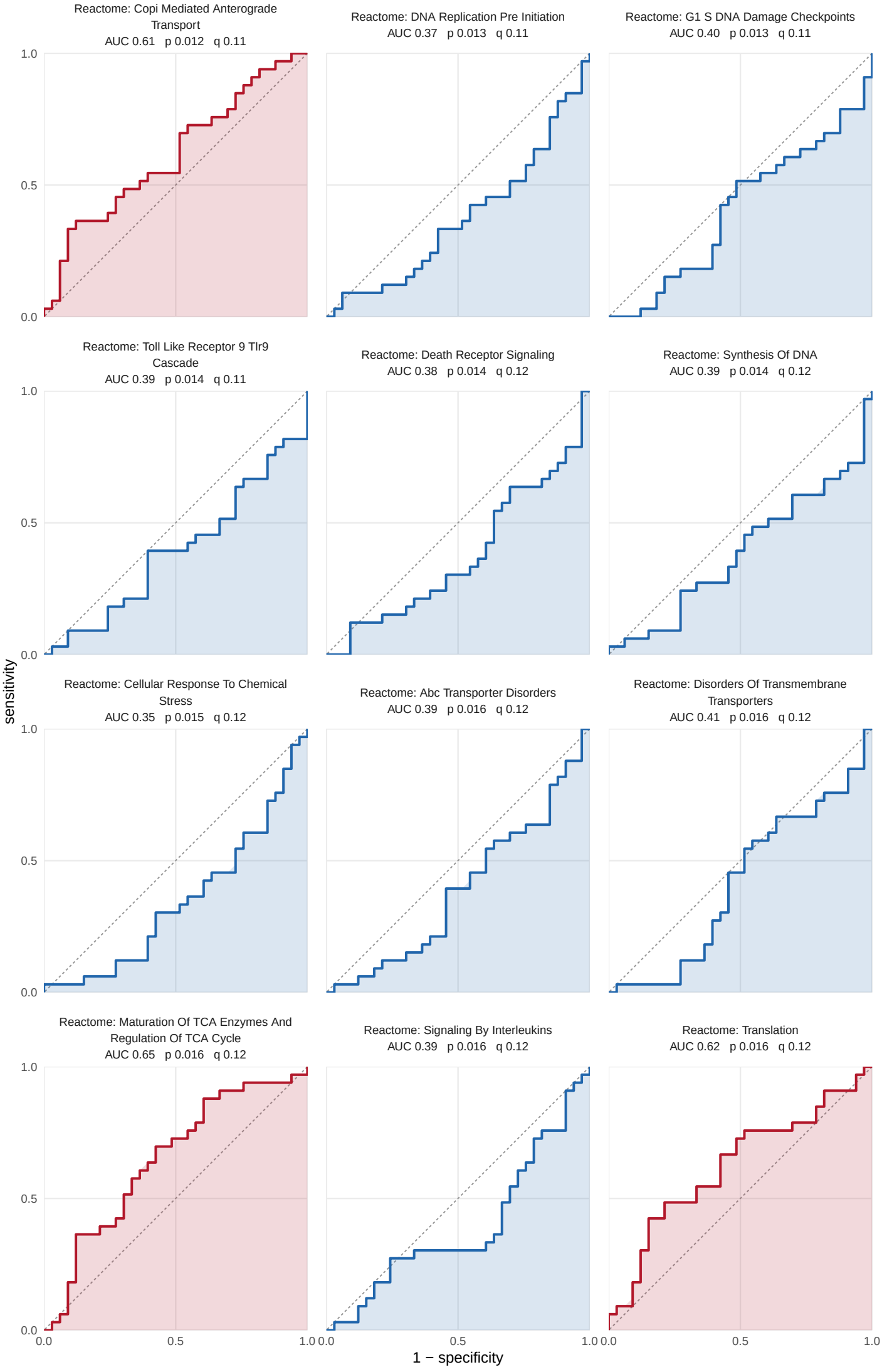


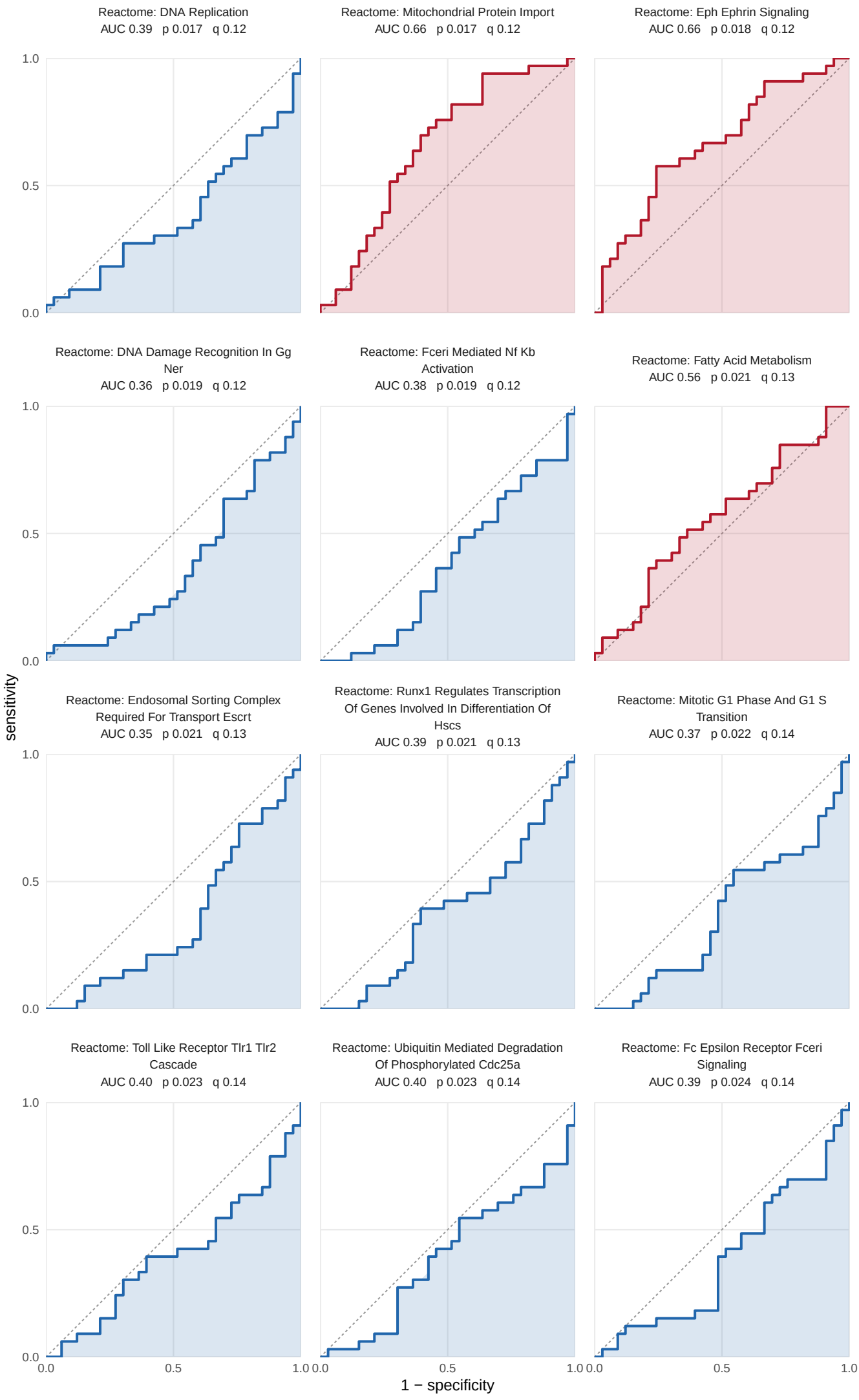

Reactome: tRNA Aminoacylation
AUC 0.69 p 2.3e−06 q 0.00064
Reactome: Mitochondrial Protein Degradation
AUC 0.69 p 3e−06 q 0.00064
Reactome: Hiv Life Cycle
AUC 0.25 p 4.6e−05 q 0.0055
Reactome: Cytosolic tRNA Aminoacylation
AUC 0.67 p 5.1e−05 q 0.0055
Reactome: Chromatin Organization
AUC 0.30 p 9.1e−05 q 0.0078
Reactome: Pyruvate Metabolism
AUC 0.72 p 0.00019 q 0.014
Reactome: Neddylation
AUC 0.32 p 0.00027 q 0.016
Reactome: Nuclear Events Mediated By Nfe2l2
AUC 0.30 p 0.00029 q 0.016
Reactome: Hcmv Late Events
AUC 0.28 p 0.00051 q 0.025
Reactome: Citric Acid Cycle TCA Cycle
AUC 0.68 p 0.00082 q 0.035
Reactome: Signaling By The B Cell Receptor Bcr
AUC 0.34 p 0.00095 q 0.037
Reactome: Aerobic Respiration And Respiratory Electron Transport
AUC 0.68 p 0.0012 q 0.039
sensitivity
1 − specificity
0.0
0.5
1.0

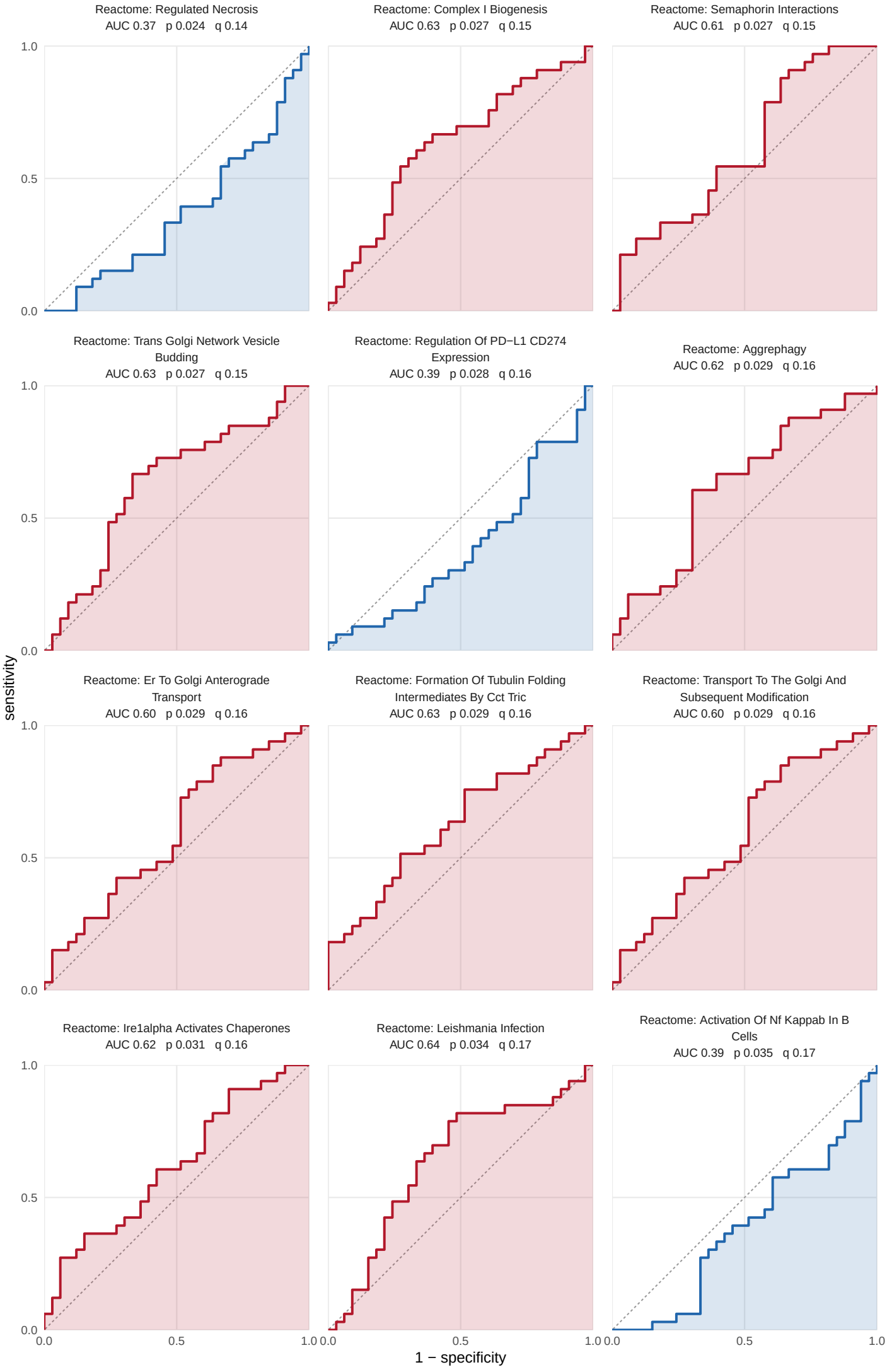

Reactome: Chromosome Maintenance
AUC 0.31 p 0.0013 q 0.039
Reactome: Mitochondrial Translation
AUC 0.67 p 0.0014 q 0.039
Reactome: Mitochondrial Translation Elongation
AUC 0.67 p 0.0014 q 0.039
Reactome: Hiv Infection
AUC 0.32 p 0.0016 q 0.04
Reactome: Regulation Of Cholesterol Biosynthesis By Srebp Srebf
AUC 0.31 p 0.0016 q 0.04
Reactome: Rho GTPase Effectors
AUC 0.70 p 0.0019 q 0.047
Reactome: Rho Gtpases Activate Wasps And Waves
AUC 0.71 p 0.0022 q 0.051
Reactome: Signaling By Retinoic Acid
AUC 0.67 p 0.0026 q 0.055
Reactome: Parasite Infection
AUC 0.71 p 0.0027 q 0.056
Reactome: Interleukin 1 Signaling
AUC 0.38 p 0.0029 q 0.057
Reactome: Formation Of The Dystrophin Glycoprotein Complex DGC
AUC 0.32 p 0.0031 q 0.058
Reactome: Antigen Processing Ubiquitination Proteasome Degradation
AUC 0.34 p 0.0046 q 0.073
sensitivity
1 − specificity
0.0
0.5
1.0

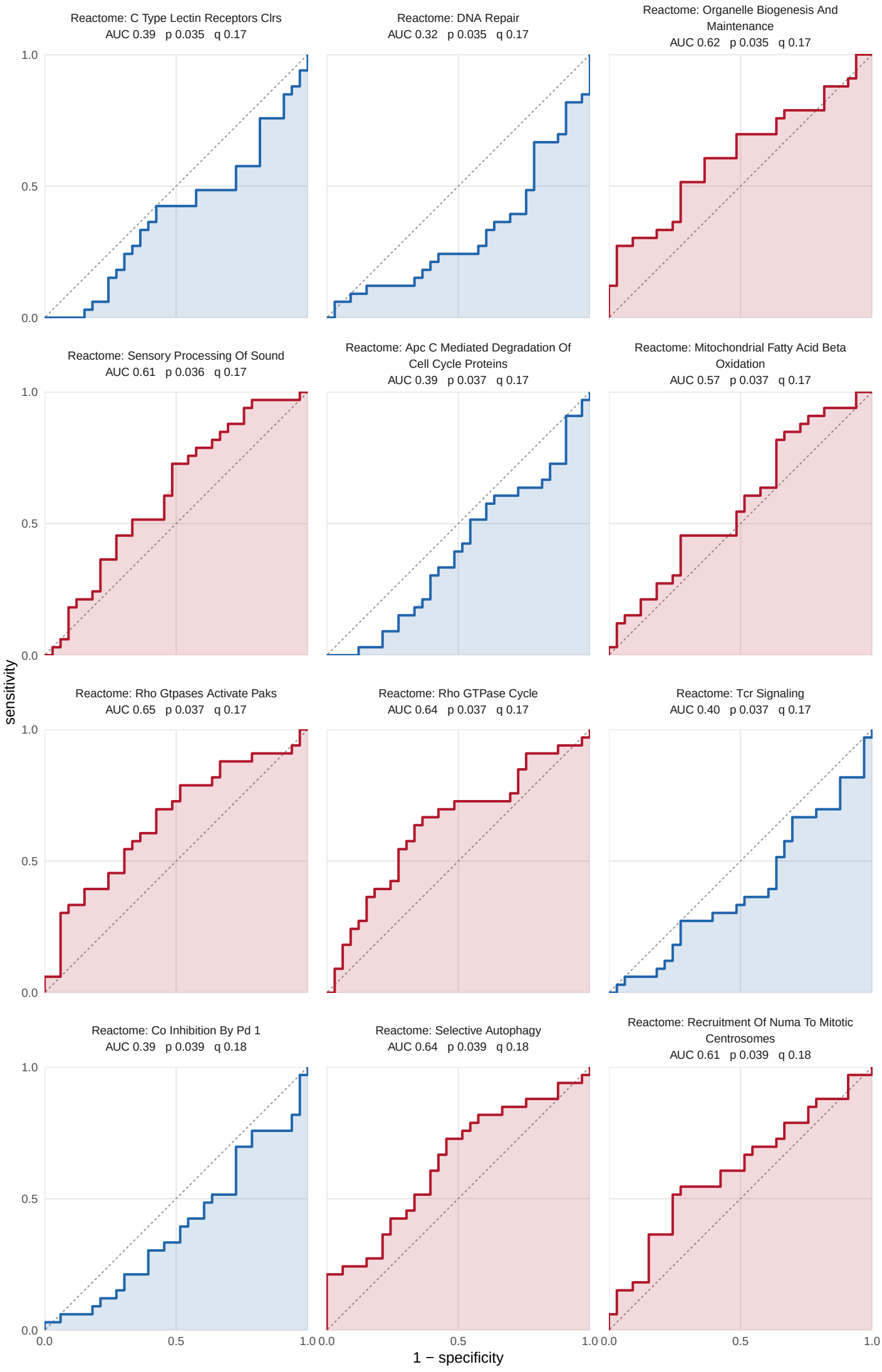

Reactome: Fcgamma Receptor Fcgr Dependent Phagocytosis
AUC 0.71 p 0.0046 q 0.073
Reactome: Glycogen Metabolism
AUC 0.66 p 0.0046 q 0.073
Reactome: Recycling Pathway Of L1
AUC 0.70 p 0.0046 q 0.073
Reactome: Nucleotide Excision Repair
AUC 0.30 p 0.0049 q 0.075
Reactome: Late Endosomal Microautophagy
AUC 0.34 p 0.0052 q 0.077
Reactome: Golgi Associated Vesicle Biogenesis
AUC 0.65 p 0.0055 q 0.079
Reactome: Global Genome Nucleotide Excision Repair Gg Ner
AUC 0.33 p 0.0062 q 0.079
Reactome: Interleukin 1 Family Signaling
AUC 0.36 p 0.0062 q 0.079
Reactome: Downstream Signaling Events Of B Cell Receptor Bcr
AUC 0.36 p 0.0066 q 0.079
Reactome: Formation Of Tc Ner Pre Incision Complex
AUC 0.33 p 0.0066 q 0.079
Reactome: Interferon Gamma Signaling
AUC 0.65 p 0.0066 q 0.079
Reactome: Budding And Maturation Of Hiv Virion
AUC 0.35 p 0.007 q 0.079
sensitivity
1 − specificity
0.0
0.5
1.0

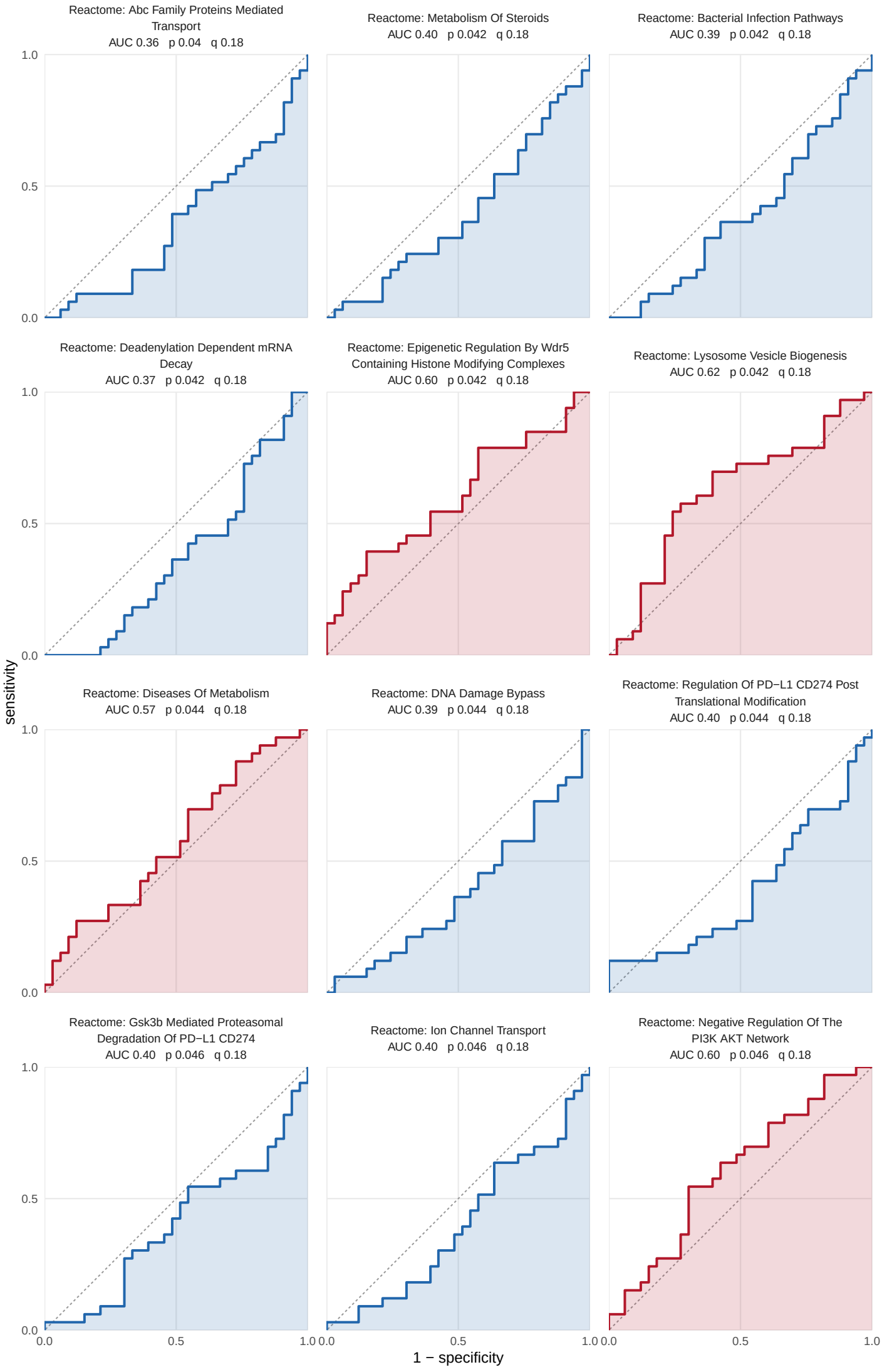

Reactome: Ephb Mediated Forward Signaling
AUC 0.68 p 0.007 q 0.079
Reactome: Transcription Coupled Nucleotide Excision Repair Tc Ner
AUC 0.31 p 0.007 q 0.079
Reactome: Clec7a Dectin 1 Signaling
AUC 0.37 p 0.0074 q 0.081
Reactome: Keap1 Nfe2l2 Pathway
AUC 0.35 p 0.0083 q 0.087
Reactome: Mitochondrial tRNA Aminoacylation
AUC 0.63 p 0.0083 q 0.087
Reactome: Cargo Trafficking To The Periciliary Membrane
AUC 0.64 p 0.0098 q 0.098
Reactome: Downstream Tcr Signaling
AUC 0.35 p 0.0098 q 0.098
Reactome: Cilium Assembly
AUC 0.63 p 0.01 q 0.098
Reactome: Respiratory Electron Transport
AUC 0.66 p 0.011 q 0.1
Reactome: Ub Specific Processing Proteases
AUC 0.38 p 0.011 q 0.1
Reactome: Transcriptional And Post Translational Regulation Of Mitf M Expression And Activity
AUC 0.60 p 0.012 q 0.11
Reactome: Antigen Presentation Folding Assembly And Peptide Loading Of Class I Mhc
AUC 0.63 p 0.012 q 0.11
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Copi Mediated Anterograde Transport
AUC 0.61 p 0.012 q 0.11
Reactome: DNA Replication Pre Initiation
AUC 0.37 p 0.013 q 0.11
Reactome: G1 S DNA Damage Checkpoints
AUC 0.40 p 0.013 q 0.11
Reactome: Toll Like Receptor 9 Tlr9 Cascade
AUC 0.39 p 0.014 q 0.11
Reactome: Death Receptor Signaling
AUC 0.38 p 0.014 q 0.12
Reactome: Synthesis Of DNA
AUC 0.39 p 0.014 q 0.12
Reactome: Cellular Response To Chemical Stress
AUC 0.35 p 0.015 q 0.12
Reactome: Abc Transporter Disorders
AUC 0.39 p 0.016 q 0.12
Reactome: Disorders Of Transmembrane Transporters
AUC 0.41 p 0.016 q 0.12
Reactome: Maturation Of TCA Enzymes And Regulation Of TCA Cycle
AUC 0.65 p 0.016 q 0.12
Reactome: Signaling By Interleukins
AUC 0.39 p 0.016 q 0.12
Reactome: Translation
AUC 0.62 p 0.016 q 0.12
sensitivity
1 − specificity
0.0
0.5
1.0

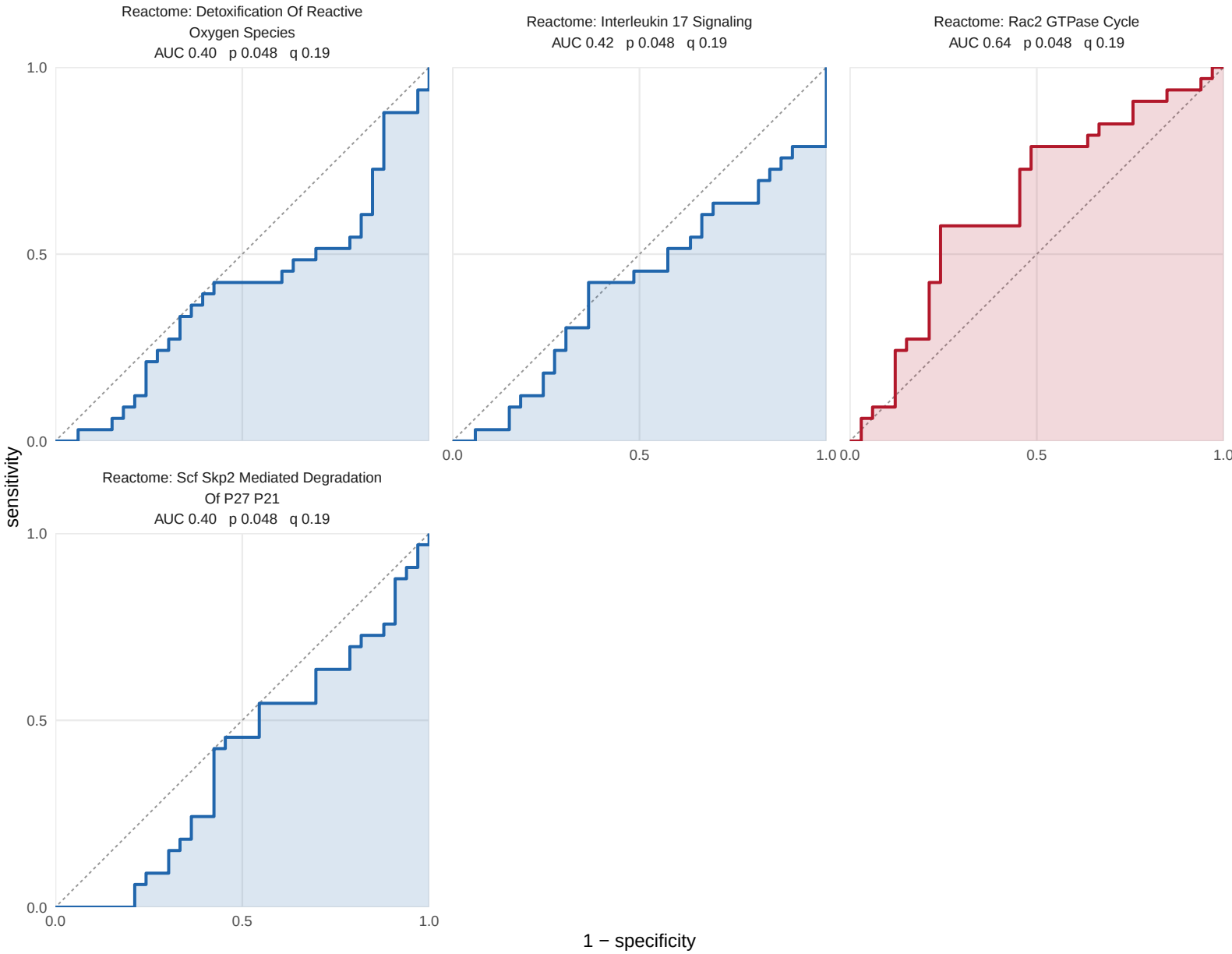

Reactome: DNA Replication
AUC 0.39 p 0.017 q 0.12
Reactome: Mitochondrial Protein Import
AUC 0.66 p 0.017 q 0.12
Reactome: Eph Ephrin Signaling
AUC 0.66 p 0.018 q 0.12
Reactome: DNA Damage Recognition In Gg Ner
AUC 0.36 p 0.019 q 0.12
Reactome: Fceri Mediated Nf Kb Activation
AUC 0.38 p 0.019 q 0.12
Reactome: Fatty Acid Metabolism
AUC 0.56 p 0.021 q 0.13
Reactome: Endosomal Sorting Complex Required For Transport Escrt
AUC 0.35 p 0.021 q 0.13
Reactome: Runx1 Regulates Transcription Of Genes Involved In Differentiation Of Hscs
AUC 0.39 p 0.021 q 0.13
Reactome: Mitotic G1 Phase And G1 S Transition
AUC 0.37 p 0.022 q 0.14
Reactome: Toll Like Receptor Tlr1 Tlr2 Cascade
AUC 0.40 p 0.023 q 0.14
Reactome: Ubiquitin Mediated Degradation Of Phosphorylated Cdc25a
AUC 0.40 p 0.023 q 0.14
Reactome: Fc Epsilon Receptor Fceri Signaling
AUC 0.39 p 0.024 q 0.14
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Regulated Necrosis
AUC 0.37 p 0.024 q 0.14
Reactome: Complex I Biogenesis
AUC 0.63 p 0.027 q 0.15
Reactome: Semaphorin Interactions
AUC 0.61 p 0.027 q 0.15
Reactome: Trans Golgi Network Vesicle Budding
AUC 0.63 p 0.027 q 0.15
Reactome: Regulation Of PD−L1 CD274 Expression
AUC 0.39 p 0.028 q 0.16
Reactome: Aggrephagy
AUC 0.62 p 0.029 q 0.16
Reactome: Er To Golgi Anterograde Transport
AUC 0.60 p 0.029 q 0.16
Reactome: Formation Of Tubulin Folding Intermediates By Cct Tric
AUC 0.63 p 0.029 q 0.16
Reactome: Transport To The Golgi And Subsequent Modification
AUC 0.60 p 0.029 q 0.16
Reactome: Ire1alpha Activates Chaperones
AUC 0.62 p 0.031 q 0.16
Reactome: Leishmania Infection
AUC 0.64 p 0.034 q 0.17
Reactome: Activation Of Nf Kappab In B Cells
AUC 0.39 p 0.035 q 0.17
sensitivity
1 − specificity
0.0
0.5
1.0

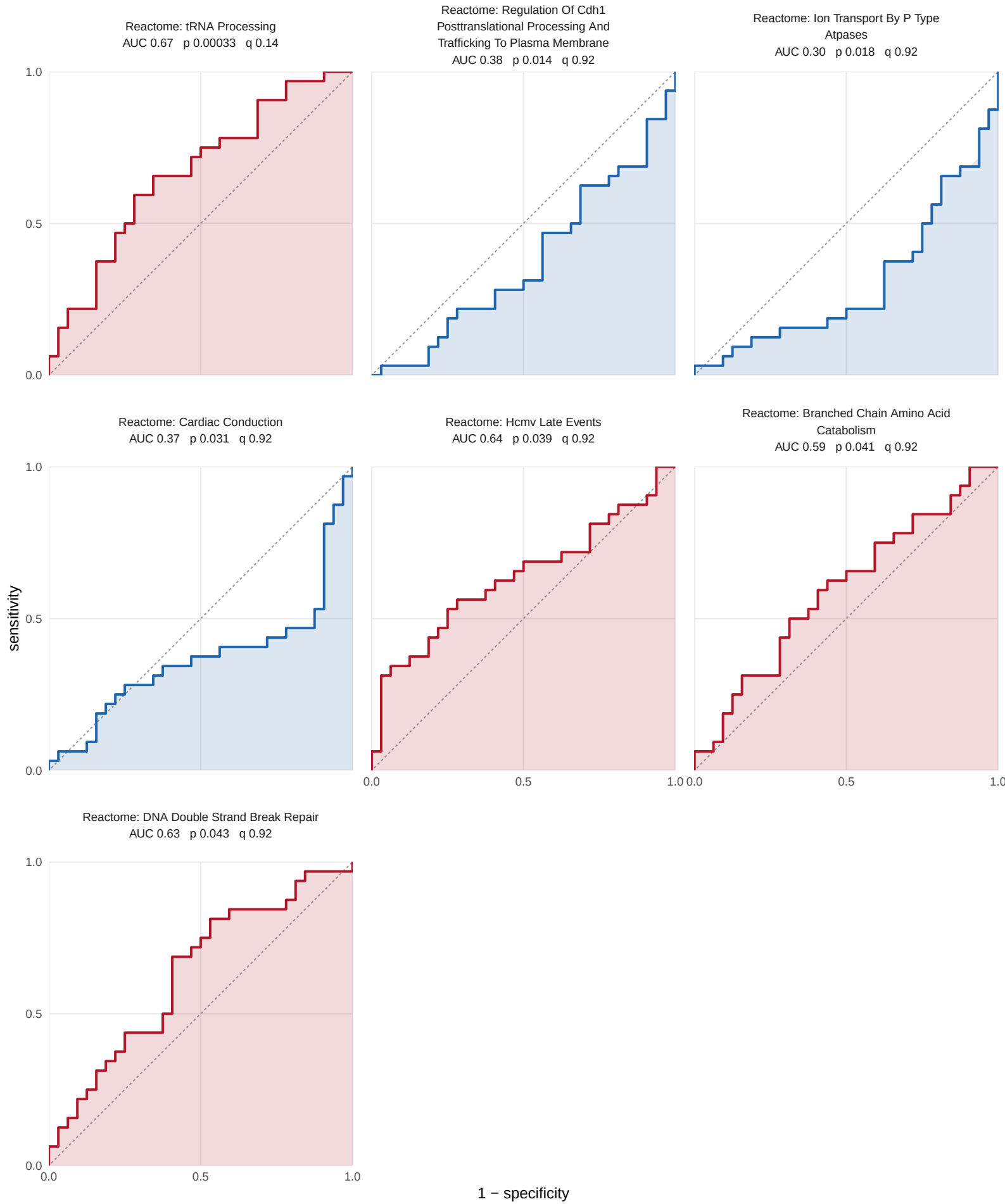

Reactome: C Type Lectin Receptors Clrs
AUC 0.39 p 0.035 q 0.17
Reactome: DNA Repair
AUC 0.32 p 0.035 q 0.17
Reactome: Organelle Biogenesis And Maintenance
AUC 0.62 p 0.035 q 0.17
Reactome: Sensory Processing Of Sound
AUC 0.61 p 0.036 q 0.17
Reactome: Apc C Mediated Degradation Of Cell Cycle Proteins
AUC 0.39 p 0.037 q 0.17
Reactome: Mitochondrial Fatty Acid Beta Oxidation
AUC 0.57 p 0.037 q 0.17
Reactome: Rho Gtpases Activate Paks
AUC 0.65 p 0.037 q 0.17
Reactome: Rho GTPase Cycle
AUC 0.64 p 0.037 q 0.17
Reactome: Tcr Signaling
AUC 0.40 p 0.037 q 0.17
Reactome: Co Inhibition By Pd 1
AUC 0.39 p 0.039 q 0.18
Reactome: Selective Autophagy
AUC 0.64 p 0.039 q 0.18
Reactome: Recruitment Of Numa To Mitotic Centrosomes
AUC 0.61 p 0.039 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Abc Family Proteins Mediated Transport
AUC 0.36 p 0.04 q 0.18
Reactome: Metabolism Of Steroids
AUC 0.40 p 0.042 q 0.18
Reactome: Bacterial Infection Pathways
AUC 0.39 p 0.042 q 0.18
Reactome: Deadenylation Dependent mRNA Decay
AUC 0.37 p 0.042 q 0.18
Reactome: Epigenetic Regulation By Wdr5 Containing Histone Modifying Complexes
AUC 0.60 p 0.042 q 0.18
Reactome: Lysosome Vesicle Biogenesis
AUC 0.62 p 0.042 q 0.18
Reactome: Diseases Of Metabolism
AUC 0.57 p 0.044 q 0.18
Reactome: DNA Damage Bypass
AUC 0.39 p 0.044 q 0.18
Reactome: Regulation Of PD−L1 CD274 Post Translational Modification
AUC 0.40 p 0.044 q 0.18
Reactome: Gsk3b Mediated Proteasomal Degradation Of PD−L1 CD274
AUC 0.40 p 0.046 q 0.18
Reactome: Ion Channel Transport
AUC 0.40 p 0.046 q 0.18
Reactome: Negative Regulation Of The PI3K AKT Network
AUC 0.60 p 0.046 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0

S7. ROC curves, Pre to post, HLRT, Reactome (page 10 of 10)



S8. ROC curves, Post BFR vs HLRT, Reactome



S9. ROC curves, Change, BFR vs HLRT, Reactome

